

3D Gaussian Splatting for Scene Editing and Completion

Qianwen Zhong
Shenzhen University
Shenzhen, Guangdong, China
2400042031@email.szu.edu.cn

Wenhui Liu
Shenzhen University
Shenzhen, Guangdong, China
liuwenhui2023@email.szu.edu.cn

Hongping Zhou
Shenzhen University
Shenzhen, Guangdong, China
2310433089@email.szu.edu.cn

Ping Lu*
State Key Laboratory of Mobile
Network and Mobile Multimedia
Technology
Shenzhen, Guangdong, China
ZTE Corporation
Shenzhen, Guangdong, China
lu.ping@zte.com.cn

Wenzhe Shi
Beijing XingYun Digital Technology
Co., Ltd
Beijing, China
ZTE Corporation
Shenzhen, Guangdong, China
shi.wenzhe@xydigit.com

Abstract

In recent years, 3D Gaussian Splatting (3DGS) has rapidly emerged as a research hotspot for next-generation 3D scene modeling and reconstruction, owing to its highly efficient differentiable representation and superior rendering performance. Building upon this foundation, 3DGS-based scene editing techniques have attracted increasing attention, offering flexible and high-quality solutions for tasks such as object insertion, deletion, deformation, and completion. This paper provides a comprehensive review of the latest research progress in 3D scene editing based on 3DGS. We first revisit the development background of 3DGS and then categorize existing editing methods from two perspectives: editing scenarios and control mechanisms. The survey covers four major categories, namely static scene editing, dynamic scene editing, semantic-driven editing, and user-interactive editing, while also summarizing mainstream scene completion approaches. Finally, we discuss the current challenges faced by this field and outline future research directions in semantic fusion, multimodal generation, and editing evaluation, aiming to provide systematic insights and references for researchers in related domains.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Computer vision; Computer vision problems; Reconstruction; Artificial intelligence; Computer vision; Computer vision tasks; Scene understanding; Artificial intelligence; Computer vision; Computer vision problems; Image segmentation.

Keywords

3D Gaussian Splatting, 3D Scene Editing, Scene Completion

*Corresponding author



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1 Introduction

3D Gaussian Splatting (3DGS) [1] has emerged in recent years as one of the most prominent revolutionary techniques in the fields of novel view synthesis and 3D reconstruction. This innovative approach introduces a 3D Gaussian representation that achieves state-of-the-art visual quality while maintaining highly competitive training efficiency. By fully leveraging the advantages of implicit radiance fields in capturing high-resolution details and the rendering efficiency of explicit radiance fields, 3DGS provides an explicit and differentiable scene representation, which has rapidly drawn extensive attention from research groups and scholars worldwide.

Current research on 3DGS is largely focused on enabling practical applications, including 3D reconstruction, augmented reality (AR), virtual reality (VR), and simultaneous localization and mapping (SLAM). Compared with the previously dominant implicit representation, namely Neural Radiance Fields (NeRF) [2], 3DGS demonstrates significantly faster deployment capabilities and shows substantial potential for future development.

In this paper, we present a systematic review of the latest advances and key contributions in the field of 3D scene editing based on 3DGS. We first introduce the technological evolution of 3D scene reconstruction and editing, with a particular emphasis on the fundamental theories underlying 3DGS and its critical role in 3D scene construction. Subsequently, we provide a comprehensive review of existing research efforts in two major areas: 3D scene editing and scene completion. Each section systematically summarizes the state-of-the-art methods, breakthroughs in key technologies, existing challenges, and promising directions for future research, aiming to establish a solid theoretical foundation for the development of this field. Finally, we discuss the potential of 3DGS across various application domains and related tasks, offering new perspectives

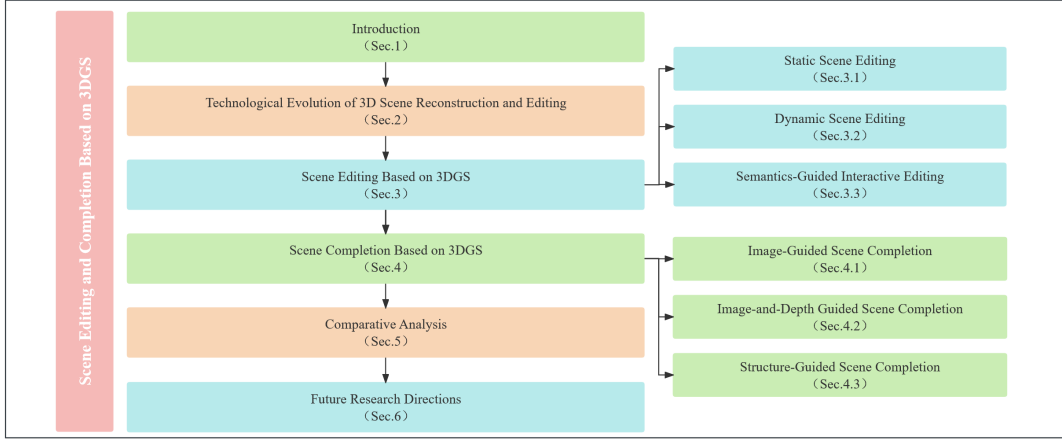


Figure 1: The overall structure of our comprehensive survey.

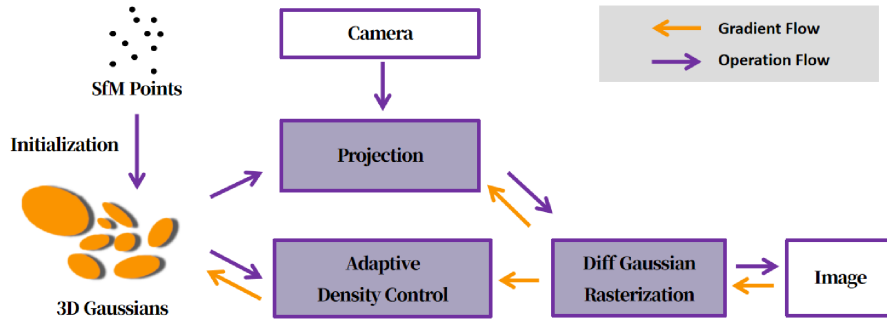


Figure 2: Schematic of the 3DGS framework.

for advancing the technology and broadening its practical impact. The overall structure of this paper is illustrated in Fig 1

Below, we provide a brief overview of the fundamental principles of 3DGS to ground the subsequent discussions. 3DGS is an efficient explicit scene representation technique that models a 3D environment using a set of anisotropic Gaussian primitives. Each primitive is parameterized by its 3D position, covariance matrix, opacity, and spherical harmonics coefficients to capture view-dependent appearance. Unlike implicit representations such as NeRF, which require dense volumetric sampling, 3DGS directly encodes the scene as a sparse collection of Gaussian distributions. During rendering, these Gaussians are projected onto the image plane and blended using a differentiable splatting operation based on alpha compositing, enabling real-time rendering while maintaining high visual fidelity.

Formally, the spatial density of a Gaussian is defined as

$$G(x) = \exp\left(-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right) \quad (1)$$

where μ is the Gaussian center and Σ is the covariance matrix that controls its spatial extent and anisotropy. The parameters of all Gaussians are jointly optimized by minimizing a photometric reconstruction loss between rendered and ground-truth images. Due to its computational efficiency and superior quality-speed trade-off, 3DGS has rapidly become a foundational representation for downstream tasks such as scene editing, completion, and semantic understanding. Beyond the fundamental formulation, the 3DGS framework is typically organized into a pipeline that consists of a forward rendering process and a backward optimization process.

Forward Process (Rendering). Given a camera pose and intrinsic parameters, each Gaussian primitive is first transformed into the camera coordinate system. The covariance matrix is projected

onto the image plane to approximate the Gaussian footprint as an ellipse. During rasterization, these 2D Gaussian ellipses are splatted onto the image plane using alpha compositing. The opacity parameter controls the blending, while spherical harmonics coefficients provide view-dependent color variations to model effects such as specular highlights. This differentiable splatting operation allows the method to produce photo-realistic images with high efficiency.

Backward Process (Optimization). To reconstruct the scene from a set of posed images, the parameters of all Gaussian primitives—including positions, covariances, opacities, and spherical harmonics coefficients—are optimized through gradient descent. A photometric reconstruction loss (e.g., mean squared error or perceptual loss) is computed between rendered images and ground-truth observations. Gradients are propagated through the differentiable splatting operation, updating Gaussian parameters to progressively align the rendered scene with the target views. This process is significantly more efficient than volumetric sampling in NeRF, as it directly updates a compact set of explicit primitives.

The overall workflow is illustrated in Figure 2. The overall pipeline thus combines explicit geometric control with differentiable rendering. The forward pass ensures real-time rendering capabilities, while the backward pass guarantees high-quality scene reconstruction. This efficient representation not only accelerates training and inference but also provides natural interfaces for downstream tasks such as scene editing, object manipulation, and semantic reasoning.

The practical impact of 3DGS is most vividly demonstrated through its transformative applications across diverse industries. In cultural heritage digitization, 3DGS enables the creation of high-fidelity, lightweight digital twins of fragile archaeological sites or artifacts, facilitating immersive online exploration and non-invasive virtual restoration. Within the realm of autonomous driving simulation, it empowers the construction of highly realistic, dynamic digital urban environments, providing a safe and efficient sandbox for rigorous algorithm testing and iteration. The technology is revolutionizing film and game production (virtual cinematography) by allowing for the real-time rendering and seamless compositing of complex virtual sets with live-action footage on set, drastically reducing production cycles and enhancing creative flexibility. Furthermore, in industrial design and augmented reality, 3DGS allows for the real-time overlay, manipulation, and collaborative review of intricate 3D product models through AR devices, streamlining processes from design evaluation and assembly guidance to remote teamwork. These applications underscore 3DGS’s pivotal role as a core enabling technology for the next generation of interactive 3D experiences.

2 Technological Evolution of 3D Scene Reconstruction and Editing

3D scene reconstruction and editing have become one of the research focuses in computer graphics and computer vision. Traditional point-based rendering methods generate 3D images using geometric primitives. These methods have improved realism and computational efficiency to some extent, but they remain limited in processing fine details in complex scenes. NeRF appeared as an implicit representation method in recent years and improved the

expressiveness and editability of 3D scene modeling, especially in detail and lighting capture. NeRF, however, still faces challenges in dynamic editing and local adjustments despite its success in generating realistic scenes. 3DGS, as an extension and improvement of NeRF, provides an explicit and efficient representation of 3D scenes. It offers greater editing flexibility and higher computational efficiency by directly modeling the geometry and lighting of each Gaussian in 3D space. This section reviews the main technical developments in 3D scene editing in recent years.

Traditional Point-Based Rendering. Point-based rendering methods aim to render a set of geometric primitives to generate visually high-quality images. Zwicker et al. [3] first proposed rendering ellipsoidal primitives that can cover multiple pixels to improve rendering precision. Later, differentiable rendering pipelines [4] [5] and anti-aliasing texture filtering [6] were introduced to further enhance the methods. These improvements allowed point-based rendering to process more complex scenes and improved both image quality and efficiency. However, most of these methods rely on predefined geometric primitives and lack flexibility and fine control over complex scene details. With the emergence of implicit representation methods, point-based rendering has gradually been replaced by approaches such as NeRF, which model the density and color fields of 3D scenes directly rather than using explicit geometric models.

NeRF-Based 3D Scene Reconstruction and Editing. NeRF has been an important breakthrough in 3D reconstruction and scene editing since its introduction. Unlike point-based rendering methods, NeRF uses an implicit volumetric rendering model [7], allowing neural networks to learn and represent the density and color fields of scenes and to achieve high visual fidelity. NeRF enables editing at the neural network level by optimizing latent codes [8], without relying on traditional geometric modeling. Approaches based on NeRF, which include latent code optimization, geometric proxy construction, and dataset updates, improve the efficiency and operability of 3D scene editing. NeRF is particularly effective for dynamic scenes with changes in shape, color, or structure. However, its dependence on dense sampling still limits efficiency and flexibility. To address these issues, researchers have started exploring more efficient scene representation methods such as 3DGS to achieve faster and more effective 3D scene editing.

3DGS-Based 3D Scene Reconstruction and Editing. 3DGS is an emerging 3D scene representation technique that combines explicit Gaussian primitives with efficient rasterization-based rendering methods, significantly improving the efficiency and flexibility of 3D scene editing. Compared to NeRF, 3DGS offers two main advantages: first, it directly models opacity, avoiding the complexity of density-to-opacity conversion present in NeRF; second, it employs rasterization-based rendering, eliminating the need for dense sampling of the entire space and reducing computational cost. By explicitly representing each 3D object with Gaussian primitives, 3DGS enables more precise and flexible object editing, including deletion, translation, rotation, scaling, as well as complex texture and geometry modifications. With ongoing advancements, 3DGS has found widespread application in scene editing and provides efficient solutions for real-time processing of dynamic scenes, thereby advancing 3D computer vision technologies [9].

Table 1: Classification of Scene Editing Methods Based on 3DGS.

Editing Category	Core Definition	Editing Objectives
Static Scene Editing	Rigid editing: editing of stationary objects with fixed geometry. Non-rigid editing: editing of stationary objects with deformable geometry.	Object insertion/deletion, rigid transformations (translation, rotation, scaling). Shape deformation, non-rigid adjustments, local geometry editing.
Dynamic Scene Editing	Editing scenes with moving or deformable objects while preserving temporal consistency.	Motion-aware modifications, dynamic object manipulation, temporal coherence.
Semantics-Guided Interactive Editing	User-driven editing guided by high-level semantic information and intuitive inputs.	Semantics-driven object replacement, appearance adjustment, interactive local editing.

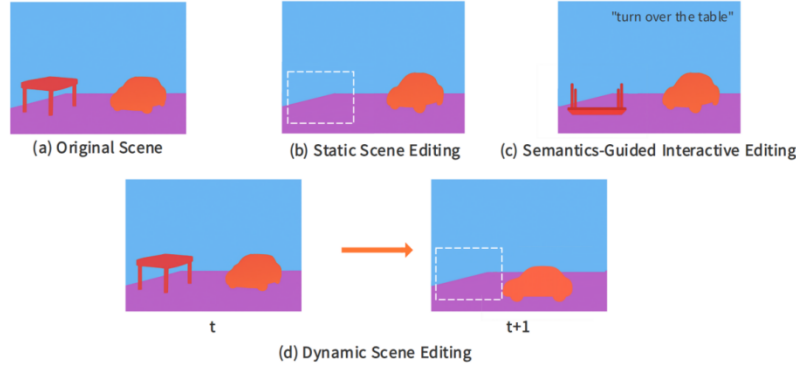


Figure 3: An overview of the three categories of 3DGS-based scene editing: (a) Original scene; (b) Static scene editing, supporting rigid and non-rigid editing; (c) Semantics-guided interactive editing, enabling editing through high-level semantic instructions; and (d) Dynamic scene editing, focusing on motion consistency across frames.

3 Scene Editing Based on 3DGS

This chapter focuses on the core tasks and implementation approaches of 3D scene editing. Based on a review of current mainstream literature and representative methods, 3DGS-based editing techniques are categorized into three types: static scene editing, dynamic scene editing, and semantics-guided interactive editing. These categories address editing scenarios and control mechanisms from both static and dynamic perspectives, while also integrating semantic understanding and user interaction. Table 1 presents a preliminary classification of these three editing methods. Fig 3 presents an overview of the three categories of scene editing based on 3DGS. The following sections will introduce and discuss each category in detail.

3.1 Static Scene Editing

In 3D scene editing, rigid and non-rigid editing are two fundamental approaches for achieving geometric transformations and shape adjustments, targeting overall structural preservation and local detail modeling of objects, respectively. Rigid editing focuses on global transformations of objects, adjusting their position and orientation through rotation, translation, or scaling. In contrast, non-rigid editing emphasizes local deformations to handle complex geometric changes in dynamic scenes. Together, these two

approaches form the essential foundation of 3D scene editing, covering a wide range of applications from static object alignment to dynamic shape modification. With recent advances in non-rigid registration techniques for shape capture and reconstruction, 3D scene editing has seen significant improvements in efficiency and accuracy, providing strong support for in-depth applications in virtual reality, augmented reality, and other computer graphics fields. However, challenges remain in processing multimodal data, modeling complex deformations, and optimizing real-time performance [10].

3.1.1 Rigid Editing. Rigid editing refers to global transformations such as rotation, translation, or uniform scaling of objects in a scene, without altering their geometric structure. It is particularly suitable for tasks that require maintaining object shape integrity while adjusting position or orientation. For example, rigid editing can efficiently achieve geometric alignment when rearranging furniture in scene reorganization or adjusting the placement of building models. Rigid editing relies on rigid registration algorithms, which aim to determine the optimal rotation and translation parameters through optimization to align the source object with the target object. Its primary applications include object localization and pose adjustment, scene alignment, and static object merging [11]. Although rigid editing has relatively low computational complexity, achieving high-precision registration remains challenging when

dealing with complex geometric structures or partially occluded objects.

Chen et al. [12] introduced GaussianEditor, which leverages Hierarchical Gaussian Splatting (HGS) for precise rigid object manipulation. HGS provides an explicit point-based structure that enables efficient object localization and editing. The method’s Gaussian semantic tracking technique allows real-time target object tracking during training, overcoming errors from static mask operations. In practice, GaussianEditor can rapidly complete object removal and replacement tasks while maintaining stability by controlling Gaussian point fluidity. HGS further refines rigid region segmentation through generation-guided optimization, ensuring high-quality object boundaries. Experiments show the method completes complex rigid editing within minutes, significantly improving efficiency and quality. Palandra et al. [13] proposed GSEdit, which enables rapid modification of the style and appearance of 3D objects while preserving their primary geometric details. This approach is particularly suitable for tasks requiring global object adjustments under multi-view consistency, such as object appearance enhancement or material replacement. Xiao et al. [14] proposed a localized Gaussian Splatting editing method that improves object replacement and insertion through global illumination consistency. By introducing the Anchor View Proposal (AVP) algorithm, the method automatically selects representative views capturing scene illumination and uses a conditional 2D diffusion model for inpainting. A coarse-to-fine optimization pipeline lifts inpainted images to 3D, ensuring illumination matching between new objects and existing scenes. Compared to traditional methods, this approach enhances texture details while maintaining global illumination consistency, producing more natural editing results.

3.1.2 Non-Rigid Editing. Non-rigid editing allows local deformations of object surfaces and is suitable for tasks requiring flexible shape adjustments, such as deformation simulation, dynamic scene reconstruction, or fine-grained object editing. Unlike rigid editing, this approach not only modifies the position of objects but also reshapes their geometry to meet scene-specific requirements. The core characteristic of non-rigid editing lies in the accurate computation of deformation fields and its high flexibility: the deformation process is typically described using parametric models or learning-based methods, enabling free-form deformations in local regions while maintaining global geometric consistency.

This editing paradigm performs particularly well in dynamic capture and repair tasks, such as human motion capture and local scene restoration in natural environments. However, several technical challenges remain significant, including the difficulty of aligning multiple surface regions in complex scenes, the trade-off between high precision and real-time performance, and the need to maintain texture and geometric consistency during deformation. Addressing these challenges while simultaneously improving editing efficiency and quality remains an important research direction [15].

In this work, we categorize non-rigid editing into two directions based on the editing objectives: geometric editing (altering poses or shapes) and appearance editing (modifying colors or textures).

In geometric editing, GaussianEditor combines Gaussian semantic tracking with HGS to provide a stable framework for non-rigid

object deformation and reconstruction. Under stochastic generation guidance, HGS effectively regulates Gaussian point attributes—especially in high-dynamic regions—achieving accurate fitting of complex non-rigid shapes. The semantic tracking module enables dynamic monitoring and adjustment of non-rigid targets, supporting seamless object removal followed by local inpainting or reinsertion, thereby enhancing generation consistency and visual quality. This approach offers an efficient solution for complex object deformation in 3D scenes. Palandra et al. [13] demonstrate that GSEdit can flexibly modify object appearance and shape using a pre-trained image diffusion model [16]. By refining local details and performing non-rigid deformations based on text instructions (e.g., adding decorative elements), it optimizes under image supervision with SDS loss to maintain multi-view consistency. However, reliance on Instruct-Pix2Pix [17] limits spatial transformation capabilities, and perspective distortions during view-consistent optimization remain challenging for dynamic non-rigid targets. Xiao et al. [14] propose the Depth-guided Inpainting SDS (DI-SDS) method, which improves non-rigid editing by enhancing geometric and texture details with strong contextual awareness. Integrating inpainting diffusion priors, it finely optimizes texture and geometry while maintaining global illumination consistency, preventing artifacts from lighting changes. This method is particularly effective for deformable objects, achieving natural and coherent non-rigid edits. Xu et al. [18] proposed TextureGS, which decouples geometry and texture to enable high-quality non-rigid texture editing. By employing a UV-mapped MLP to project 3D Gaussians into a unified 2D texture space, it supports separate modeling of structure and appearance. Enhanced with symmetric Chamfer distance and cycle-consistency losses, the method achieves real-time rendering at 58 FPS on DTU with strong multi-view consistency (PSNR 30.03, LPIPS 0.0135). While limitations remain in multi-object scenes and thin structures, TextureGS provides an effective decoupling framework for virtual production applications.

3.2 Dynamic Scene Editing

Dynamic scene editing techniques aim to enable flexible editing of 3D scenes with temporal variations while preserving high-quality rendering effects. Unlike static scenes, dynamic scenes contain geometry and texture information that changes over time, posing greater challenges for modeling and editing using 3DGS. In recent years, researchers have introduced temporal modeling mechanisms to achieve accurate modeling and editing of moving targets, including dynamic object insertion, replacement, occlusion handling, and spatiotemporal consistency adjustments. These approaches fully leverage the differentiable rendering capabilities and compact representations of 3DGS, providing an efficient and flexible solution for tasks such as film visual effects, virtual reality, and video scene understanding.

Wu et al. [19] proposed 4DGS, which extends 3D Gaussians into the temporal dimension via a Gaussian deformation field network. Using a “canonical 3D Gaussians + 4D neural voxels” representation, it predicts time-varying Gaussian parameters with lower storage complexity than frame-by-frame approaches. The method enables long-term dynamic modeling and real-time rendering, providing a scalable path for dynamic neural rendering. Luiten et al. [20]

introduced a 6-DOF tracking approach using dynamic Gaussian particles with physical motion priors (e.g., local rigidity). Through differentiable rendering and rigidity constraints, it optimizes directly from images without optical flow, ensuring physically consistent deformation and accurate tracking. The method supports dynamic object manipulation and multi-view synthesis, offering a novel solution for real-time dynamic scene modeling. Shin et al. [21] proposed Video-3DGS, a two-stage framework enhancing temporal consistency in zero-shot video editing. It initializes foreground/background 3D Gaussians using masked point clouds, then applies temporal constraints and recursive refinement to the video diffusion model. This improves reconstruction quality (PSNR) and accelerates training, demonstrating strong potential for dynamic video editing and AR/VR applications. Sun et al. [22] introduced Video Gaussian Representation (VGR), an explicit video representation embedding content into 3D Gaussians with time-dependent motion attributes. By lifting 2D priors (optical flow, depth) into compact 3D representations, VGR enables efficient video processing for tasks including tracking, depth refinement, motion editing, and novel view synthesis, providing a versatile framework for complex video manipulation. Ji et al. [23] proposed Segment Any 4D Gaussians (SA4D), a framework for fast segmentation in 4D Gaussian distributions. It introduces a Temporal Identity Feature Field (TIFS) to address Gaussian drift and employs a Gaussian deformation field to track object motion while maintaining a canonical 3D representation. Using video object tracking masks and TIFS-enhanced features, SA4D achieves high-quality segmentation and supports editing tasks like object deletion and recoloring for 4D scene manipulation.

3.3 Semantics-Guided Interactive Editing

Semantics-guided interactive editing leverages high-level semantic information—such as natural language instructions, semantic labels, or learned feature embeddings—to enable intuitive and controllable 3D scene manipulation. Unlike traditional rigid or non-rigid editing approaches that primarily focus on geometric transformations, this paradigm integrates semantic understanding with interactive user inputs, allowing users to specify editing objectives more naturally while maintaining fine-grained control over local modifications. For instance, tasks such as “replace the sofa with a chair” or “set the grass on fire” can be achieved by mapping semantic instructions to corresponding changes in geometry, appearance, or scene composition, while interactive tools like brush strokes, drag-and-drop operations, or gesture-based controls allow precise adjustments with real-time feedback. By combining semantic reasoning with user-driven interaction, this approach significantly enhances flexibility, accessibility, and efficiency in 3D Gaussian Splatting-based scene editing.

Recent research has explored combining 3DGS with multimodal semantic guidance for more precise control. Some methods directly use semantic segmentation maps or text embeddings to identify the target objects or regions to be edited. Others integrate large language models (LLMs) or vision-language models to parse complex editing instructions, thereby enabling more expressive editing operations. This paradigm bridges low-level geometric manipulation and high-level user intent, offering an efficient solution for

editing large-scale or semantically complex scenes. However, several technical challenges remain. First, the ambiguity in language instructions or semantic labels can lead to editing errors, especially in cluttered or occluded regions. Second, ensuring spatial-temporal consistency during semantic-driven modifications is non-trivial, particularly for dynamic scenes. Finally, the balance between semantic interpretability and the fine-grained control of geometry and appearance remains an open problem. Future work will likely focus on integrating stronger multimodal priors, improving semantic localization accuracy, and exploring real-time feedback mechanisms to further enhance the usability and robustness of semantics-driven editing.

Chen et al. [12] proposed GaussianEditor, a semantics-driven editing framework based on 3D Gaussian Splatting. The core idea employs Gaussian Semantic Tracing and HGS to achieve efficient 3D scene editing. It dynamically back-projects 2D segmentation masks to assign semantic labels to 3D Gaussians and continuously tracks target regions during training, overcoming control failure from static masks. HGS constrains Gaussian mobility across generation stages to suppress diffusion model instability, yielding detailed edits. A key limitation is its reliance on 2D diffusion models, which restricts performance with complex multi-object prompts. Xiao et al. [14] introduced a semantics-driven scene editing method using 3D Gaussian Splatting. Its innovation combines text-instruction guided local reconstruction with scene illumination context optimization, enabling global illumination consistency in object replacement and insertion. The method achieves a CLIP Score of 0.7817, improving over GaussianEditor by 4.56%, with each edit taking 239 seconds. However, viewpoint bias in diffusion models causes multi-view inconsistency, and it cannot handle instructions involving topological changes. Zhang et al. [24] proposed 3DitScene, a semantics-driven 3D scene editing framework based on language-guided disentangled Gaussian Splatting. It distills CLIP [25] language features into 3D Gaussian representations to build semantic-geometric joint embeddings, enabling cross-dimensional editing from 2D to 3D. The pipeline includes optimizing 3D Gaussians, establishing semantic correlations via CLIP features, and achieving object-level control through disentangled Gaussians. A limitation is incomplete semantic disentanglement for occluded or texture-similar objects. Palandra et al. [13] proposed GSEdit, a user-interactive editing framework based on 3D Gaussian Splatting. It combines text-driven diffusion models (Instruct-Pix2Pix) with efficient gradient backpropagation to enable real-time 3D object editing without precomputed semantic masks. Users input natural language commands, with edits averaging 239 seconds, and can adjust parameters via interactive sliders. While introducing lightweight language-driven editing, it requires integration with explicit semantic control methods for enhanced precision. Huang et al. [26] proposed Sparse-Controlled Gaussian Splatting (SC-GS), a dynamic 3DGS modeling framework based on sparse control points. It constructs a motion subspace using control points and an MLP, adaptively adjusting distribution to express regional motion complexity. An ARAP loss maintains local rigidity while decoupling motion and appearance representations, achieving real-time dynamic view synthesis at 30 fps with geometric consistency. The framework supports natural motion generation beyond training sets, suitable for VR and animation applications. Qu et al. [27] proposed Drag Your Gaussian

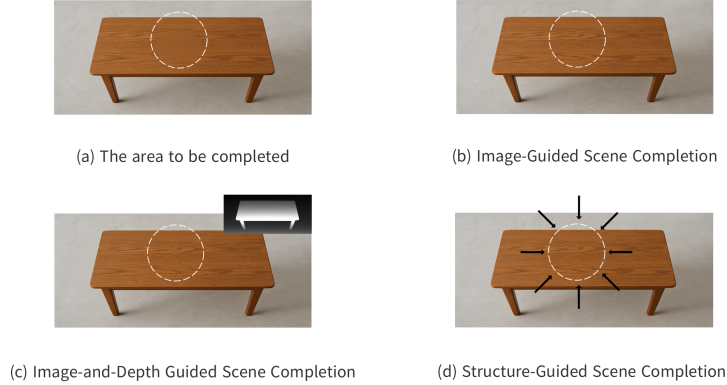


Figure 4: An overview of the three categories of 3DGS-based scene completion: (a) The scene of the area to be completed; (b) Image-Guided Scene Completion; (c) Image-and-Depth Guided Scene Completion; and (d) Structure-Guided Scene Completion.

(DYG), implementing high-quality local 3D scene editing through hybrid explicit-implicit representation and improved Score Distillation. The geometric editing layer employs multi-resolution tri-plane encoders and region-specific decoders to map user masks into implicit fields, while Drag-SDS loss enhances local precision and consistency. DYG outperforms baselines in subjective metrics with 10-minute average runtime, establishing a complete interactive workflow despite lacking real-time performance. Yan et al. [28] proposed Dragen3D, a drag-based control framework for 3D Gaussian generation and editing, and SuperSplat, a web-based editing tool. Dragen3D uses a seed-point-driven strategy with Anchor-GS VAE architecture to ensure multi-view consistency and intuitive interaction through sparse seed point manipulation. SuperSplat is an open-source browser tool supporting PLY format Gaussian clouds with full editing capabilities via WebGL rendering, providing a lightweight online solution for 3D scene manipulation.

4 Scene Completion Based on 3DGS

Removing pre-captured 3D objects from scenes is a complex and critical task in the field of 3D vision, attracting significant attention in computer vision and computer graphics, especially for applications in virtual reality and content generation. Compared to its 2D counterpart—image inpainting—3D object removal is more challenging due to the need for complex geometric completion. Moreover, the choice of 3D representation plays a crucial role in the model’s performance and rendering quality. 3DGS leverages the distribution properties of Gaussian functions to effectively represent and reconstruct the geometric structure of scenes, offering a novel approach to handling complex geometric completion tasks. However, despite its capabilities, 3DGS still faces several pressing challenges. These include the complex interplay between geometry and semantics, difficulties in reconstructing clear boundaries and structural details, insufficient geometric accuracy, and maintaining texture consistency. Addressing and improving 3DGS methods to overcome these challenges is therefore essential for advancing its practical applications.

To systematically analyze existing scene completion approaches, we categorize current methods into three main groups based on the types of guidance they leverage: image-guided, image-and-depth guided, and structure-guided completion. Each category differs in its core assumptions, available priors, and completion objectives. Table 2 provides a concise overview of these categories, highlighting their definitions and primary goals. Fig 4 illustrates the three types of complete process for scene completion based on 3DGS.

4.1 Image-Guided Scene Completion

Huang et al. [29] proposed Point’n Move, which uses Gaussian splatting radiance fields and a self-prompted segmentation algorithm to map 2D points to 3D masks, enabling efficient inpainting and real-time editing. Mask refinement provides high-quality initialization for scene completion with low computational cost, though it focuses on geometric repair without addressing lighting and texture, potentially causing darker completion results. This interactive approach offers a novel solution for real-time scene completion. Ye et al. [30] introduced Gaussian Grouping, a joint reconstruction and segmentation method based on 3D Gaussians. It enhances Gaussian identity encoding to group arbitrary content in open-world scenes by integrating SAM’s 2D mask prediction with 3D spatial consistency. Compared to implicit representations, it supports high-quality, fine-grained reconstruction and segmentation. Based on this, a local Gaussian editing scheme is proposed, balancing visual quality and efficiency for scene completion. Wang et al. [31] proposed GScream, which optimizes Gaussian primitive positions and texture propagation for efficient object removal and scene completion. The method uses monocular depth-guided training to enhance geometric consistency and cross-attention feature regularization to improve texture coherence. By facilitating information interaction between visible and occluded regions, GScream achieves high-quality texture restoration with improved training and rendering efficiency, outperforming NeRF-based methods in novel view synthesis and repair consistency. Shen et al. [32] proposed FlashSplat, a globally optimal 3D Gaussian Splatting segmentation solver

Table 2: Classification of Scene Completion Methods Based on 3DGS.

Completion Category	Core Definition	Completion Objectives
Image-Guided	Utilizes RGB images or user-provided visual cues to fill in missing or occluded regions based on appearance priors.	Complete occluded or missing regions based on texture similarity and visual plausibility.
Image-and-Depth Guided	Combines RGB information with depth maps to enhance geometric consistency and reduce reconstruction ambiguities.	Restore fine-grained geometry and improve spatial accuracy in the completed scene.
Structure-Guided	Leverages high-level semantic or structural priors to complete large missing areas.	Generate structurally consistent content, ensuring global scene integrity and semantic coherence.
Completion Category	Core Definition	Completion Objectives

that precisely lifts 2D masks into 3D space. Treating rendering as a linear function of Gaussian labels, it applies a closed-form linear program for one-step optimization, achieving $\sim 50\times$ speedup over traditional methods. Leveraging alpha blending robustness, it excels in object removal, inpainting, and novel view mask rendering, leading in efficiency and accuracy for 3D scene understanding and editing tasks.

4.2 Image-and-Depth Guided Scene Completion

Liu et al. [33] proposed InFusion, which combines diffusion priors with depth completion for efficient 3D scene inpainting and object insertion. Using an image-conditioned depth completion model to generate initial point positions aligned with missing regions, it optimizes 3D Gaussian rendering attributes. The method achieves $\sim 20\times$ speedup over existing approaches with strong restoration quality, though it has limitations in handling lighting variations and complex structural restoration. Xu et al. [34] introduced StreetUnveiler, a hard-label semantic 2D Gaussian (2DGS) approach for reconstructing empty street scenes from vehicular videos. By removing static object representations and using alpha maps to segment scenes, it minimizes repair areas. A temporal reversal repair framework leverages long trajectory observations to enhance consistency. The method effectively reconstructs empty street representations, supporting free-viewpoint rendering for autonomous driving applications. Yan et al. [35] proposed a structure-guided 3D scene completion and editing method based on direct geometric manipulation of Gaussian splatting. Using instance segmentation and CLIP-based text alignment, it performs object addition, relocation, recoloring, replacement, and deletion by directly adjusting 3D Gaussian attributes. The method achieves real-time editing (<0.5 seconds) with high accuracy, though cross-view consistency for extreme viewpoints remains a challenge. Huang et al. [36] proposed 3DGIC, a depth-guided cross-view consistency framework for high-precision 3D scene completion. Integrating depth estimation with 3D Gaussian splatting, it uses depth maps as spatial priors and cross-view constraints to ensure geometric coherence. The method reduces 3D geometric error by 42% compared to RGB-only inpainting and supports real-time editing, though it remains sensitive to initial depth map quality.

4.3 Structure-Guided Scene Completion

Pardoel et al. [37] proposed a structure-guided 3D scene completion method that directly manipulates 3D Gaussians to reconstruct missing regions, avoiding blurring from traditional multi-view inpainting. It extracts surface meshes to guide geometry, then uses PatchMatch-based [38] non-local similarity matching in 3D space to copy optimal Gaussian patches to target areas. Multi-scale rendering optimization aligns details. The method excels in geometric and texture consistency for structured scenes but faces limitations on non-rigid surfaces due to mesh dependency and lacks semantic priors for object-level plausibility. Yan et al. [35] proposed 3DSceneEditor, a structure-guided framework that directly manipulates 3D Gaussian Splatting properties for real-time scene completion and editing. It analyzes and optimizes local Gaussian distributions to reconstruct missing regions while maintaining geometric coherence. The system achieves sub-second editing latency ($<0.5s$) and higher accuracy by avoiding 2D-to-3D projection errors. Key components include semantic instance segmentation, CLIP-based prompt alignment, and direct Gaussian attribute control for object operations. Limitations remain in extreme occlusion scenarios requiring manual intervention. This work sets a benchmark for geometry-prioritized 3D scene customization, especially suited for architectural and industrial applications.

5 Comparative Analysis

5.1 Comparative Analysis of Methods

Tables 3 and 4 will respectively present the analysis and summary of the editing methods and completion methods mentioned in this article.

From the table 3, Most methods focus on sub-object level editing and show strong performance when leveraging camera parameters or geometric priors. Among them, 4DGS [19] and SC-GS [26] achieve the best overall results, while methods relying on additional geometric inputs, such as DYG [27], demonstrate higher adaptability to diverse datasets.

The analysis in Table 4 reveals a fundamental trade-off between guidance specificity and computational efficiency. Methods relying solely on semantic cues (e.g., [29, 30]) achieve real-time performance by operating as efficient 2D-to-3D mask lifters; however, this efficiency comes at the cost of volumetric plausibility, as they propagate 2D masks without resolving inherent ambiguities into

Table 3: Comparison of representative 3DGS-based scene editing methods in terms of editing granularity, input modalities, datasets, and quantitative performance.

Method	Granularity	Input Type	Dataset
TextureGS [18]	Object Level	RGB + Cam Poses	DTU [39]
4DGS [19]	Sub-Object Level	RGB Video + Cam Params	Plenoptic Video [40]
Luiten et al. [20]	Sub-Object Level	RGB Video + Cam Params	PanopticSports [41]
SC-GS [26]	Sub-Object Level	RGB + Cam Poses	Plenoptic Video [40]
			DAVIS-2023 [42]
DYG [27]	Sub-Object Level	RGB + Geo-Inter Labels	Mip-NeRF360 [43]
			Custom Synthetic [44]
Dragen3D [28]	Sub-Object Level	RGB + Depth / Point Cloud	Objaverse [45]
			GSO [46]

Table 4: summarizes representative 3DGS-based scene completion methods in terms of guidance type, input modalities, feature fusion strategy, and real-time capability. RT (Real-time) indicates that the method supports real-time processing or rendering, typically achieving interactive frame rates (≥ 30 FPS).

Method	Guidance Type	Input Modalities	Feature Fusion Strategy	RT	Dataset
Point’n Move [29]	Semantic-guided	RGB + 2DGS	Separate optimization	✓	SPIIn-NeRF [49]
Gaussian Grouping [30]	Semantic-guided	RGB + SAM masks	Separate optimization	✓	Mip-NeRF 360 [43]
GScream [31]	Hybrid (Depth + Geometric Optimization)	RGB + Depth	Cross-attention	×	SPIIn-NeRF [49] IBRNet [50]
FlashSplat [32]	Semantic-guided	RGB + SAM masks	No fusion	✓	Mip-NeRF 360 [43] T&T [51] LLFF [47] Instruct-NeRF2NeRF [52] LERF [53] NVOS [54]
InFusion [33]	Hybrid (Image + Depth)	RGB + Depth	Separate optimization	✓	SceneFlow [55] Mip-NeRF 360 [43] SPIIn-NeRF [49]
StreetUnveiler [34]	Semantic-guided	RGB + 2DGS	Separate optimization	×	Waymo [56]
3dsceneeditor [35]	Hybrid (Semantic + Textual)	RGB + CLIP	Separate optimization	✓	ScanNet++ [48]
3DGIC [36]	Hybrid (Semantic + Textual)	RGB + CLIP	Separate optimization	✓	SPIIn-NeRF [49] Mip-NeRF 360 [43] InNeRF360 [57] LERF [53]

coherent 3D structures. Conversely, hybrid methods (e.g., [31, 33]) incorporate additional modalities like depth or text to enhance geometric and semantic control. Yet, their fusion strategies often based on cross-attention or separate processing, struggle with a feature alignment crisis: misalignments between geometric edges, semantic boundaries, and linguistic descriptions lead to compounded errors, such as texture-geometry mismatches, which increase computational overhead and often preclude real-time execution.

This trade-off underscores a deeper technical challenge beyond fusion efficiency: the need for a unified, 3D-native representation where multi-modal cues are inherently grounded and reconciled in 3D space, rather than forcibly merged in post-hoc processing. Future work must therefore address this representational gap to achieve scene completion that is both semantically and geometrically precise, while remaining computationally tractable.

Table 5: Overview of representative 3D scene datasets commonly used for 3DGS-based editing and completion methods. The table summarizes the venue, number of scenes, average number of views per scene, and characteristics of each dataset, including capture setup, resolution, and scene complexity. These datasets cover a variety of scenarios, from synthetic objects on simple backgrounds to real-world dynamic multi-view scenes.

Dataset	Venue	#Scenes	#Views (Avg.)	Description	Associated Methods
DTU [39]	[CVPR’14]	80	343	Each scene consists of 49 or 64 accurate camera positions and reference structured light scans.	TextureGS [18]
Plenoptic Video [40]	[CVPR’22]	6	19	Mobile multi-view capture using 21 synchronized GoPro cameras, capturing challenging dynamic scenes with view-dependent effects, motion, shadows, volumetric behaviors, and lighting variation.	4DGS [19] SC-GS [26]
PanopticSports [41]	[arXiv 2024]	6	31	Multi-view dynamic human-object scenes with static background; hemispherical camera rig; 640×360 resolution; dense point cloud from 10 depth cameras.	Luiten et al. [20]
LLFF [47]	[TOG’19]	32	25	Using the COLMAP structure from motion implementation to compute poses for real images.	FlashSplat [32]
Mip-NeRF360 [43]	[CVPR’22]	9	215	It consists of 360-degree and outdoor scenes.	DYG [27] Gaussian Grouping [30] FlashSplat [32] InFusion [33] 3DGIC [36]
ScanNet++ [48]	[ICCV’23]	460	608	280,000 captured DSLR images, and over 3.7M iPhone RGBD frames.	3dsceneeditor [35]
LERF [53]	[ICCV’23]	20	30	Open-vocabulary neural representations enabling language-driven localization and interaction in real-world indoor scenes.	FlashSplat [32] 3DGIC [36]

5.2 Comparative Analysis of Datasets

To facilitate the evaluation of 3DGS-based scene editing and completion methods, a wide range of datasets have been employed in recent studies. These datasets differ in terms of scale, acquisition setup, and scene diversity, ranging from controlled synthetic benchmarks to large-scale real-world captures. Table 5 provides a consolidated overview of representative datasets, summarizing their venues, number of scenes, average views, and distinctive characteristics. By covering synthetic, indoor, outdoor, and dynamic scenarios, these datasets establish comprehensive testbeds for analyzing the robustness, generalization, and scalability of 3DGS techniques.

Table 5 illustrates the diversity of datasets used in 3DGS research. Synthetic datasets, such as DTU, provide high-quality images with precise camera calibration, suitable for evaluating reconstruction and editing accuracy. Dynamic or real-world datasets, such as Plenoptic Video and PanopticSports, introduce challenges like motion, complex lighting, shadows, and volumetric effects, essential for assessing robustness. Large-scale RGB-D datasets, such as ScanNet++, offer dense multi-view captures with depth information, supporting fine-grained geometry reconstruction.

Future datasets for 3D scene editing and completion should combine high-quality geometry, multi-view imagery, dynamic elements, diverse scene types, and annotated semantics, while also providing sufficient scale for training and evaluation. Such datasets would

enable more comprehensive benchmarking and foster the development of robust, generalizable 3DGS methods.

6 Conclusion and Future Research Directions

This survey has examined the trajectory of 3D Gaussian Splatting in scene editing and completion, highlighting a core representational trade-off between semantic fidelity and computational efficiency. The analysis reveals that current methods—whether semantics-driven or hybrid—remain constrained by post-hoc feature alignment, limiting both geometric coherence and real-time applicability. Looking ahead, the convergence of 3DGS with AIGC and autonomous perception signals a paradigm shift: from isolated reconstruction tasks toward unified, 3D-native representations that inherently reconcile multi-modal cues. Addressing this representational gap is not merely an optimization problem but a foundational step toward scalable, semantically controllable, and physically plausible scene understanding. Future progress will depend on datasets that bridge synthetic precision and real-world complexity, as well as on architectures that embed semantics directly into 3D structure. Within this evolving landscape, 3DGS stands poised to transition from a powerful rendering primitive to a core engine for intelligent content generation and embodied perception.

The rise of Artificial Intelligence Generated Content (AIGC) marks a fundamental shift in digital media production. AIGC refers to content created or significantly altered by AI systems, spanning

domains such as computer vision, natural language processing, and machine learning. Its significance lies in the ability to automate and scale the generation of realistic and customizable digital assets, offering transformative potential across entertainment, education, and beyond. Within this landscape, the explicit representation and real-time rendering capabilities of 3DGS provide an especially promising technical foundation. The method’s differentiable rendering pipeline aligns seamlessly with AIGC’s demand for high-fidelity, interactive, and editable 3D content, enabling applications from real-time virtual humans to interactive scene manipulation. Recent studies illustrate this synergy—for instance, by integrating 3DGS with generative models for text-to-3D object synthesis, dynamic avatar creation from monocular video, and open-world scene segmentation and editing.

Autonomous driving systems rely on the accurate and real-time perception of dynamic environments to navigate safely. Vehicles equipped with multimodal sensors must continuously reconstruct their surroundings, identifying both static structures and moving objects while interpreting spatial relationships and motion patterns. This task is particularly challenging in dense urban settings where scenes are complex and rapidly evolving. 3D Gaussian Splatting offers a compelling solution for reconstructing driving scenes by unifying sparse and irregular sensor data—such as LiDAR point clouds—into a coherent, continuous 3D representation. Its ability to handle varying point densities and efficiently model both static backgrounds and dynamic objects makes it well-suited for real-time scene understanding in autonomous systems. Although still an emerging area, recent research demonstrates that 3DGS-based approaches can achieve superior reconstruction quality and temporal consistency compared to existing methods, pointing toward its potential as a key component in next-generation autonomous perception pipelines.

Although 3DGS has demonstrated strong representational capacity and rendering efficiency in 3D scene modeling and editing, existing methods still face significant challenges in accuracy, generalization, and semantic control capabilities. Future research is expected to advance 3DGS techniques for scene editing and completion from the following perspectives:

Integration of semantic understanding and high-level control. Most current 3DGS editing approaches focus on geometric and appearance-level processing, lacking in-depth understanding of scene semantic structures. Future work may incorporate high-level semantic modules such as semantic segmentation, object detection, and language modeling to enable high-level control based on natural language or instance categories, thereby enhancing the intelligence and controllability of editing. For example, integrating multi-modal pre-trained models like CLIP can facilitate semantic-guided editing by mapping “language to Gaussian representations,” which can also be leveraged for semantic completion and reconstruction of missing regions.

Efficient optimization and real-time editing systems. While 3DGS inherently offers high rendering efficiency, editing and completion of large-scale scenes still encounter challenges such as storage redundancy and high computational cost. Future research should focus on compression and optimization of Gaussian representations, real-time interactive editing frameworks, and incremental optimization algorithms to further reduce latency and improve

system responsiveness, which is especially critical in applications requiring low-latency feedback such as VR and AR.

Multimodal fusion and generative capability expansion. Incorporating multi-modal data, including images, language, depth, and point clouds, into the 3DGS modeling pipeline is expected to yield more robust and generalized 3D modeling and editing capabilities. The integration of generative AI techniques could extend 3DGS from reconstructing existing scenes to fully generative modeling, supporting few-shot or zero-shot structural completion and style editing, thereby progressing toward cognitively capable 3D content generation systems.

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References

- [1] Kerbl B, Kopanas G, Leimkühler T, *et al.* 3d gaussian splatting for real-time radiance field rendering[J]. *ACM Trans. Graph.*, 2023, 42(4): 139:1-139:14.
- [2] Mildenhall B, Srinivasan P P, Tancik M, *et al.* Nerf: Representing scenes as neural radiance fields for view synthesis[J]. *Communications of the ACM*, 2021, 65(1): 99-106.
- [3] Zwicker M, Pfister H, Van Baar J, *et al.* Surface splatting[C]//*Proceedings of the 28th annual conference on Computer graphics and interactive techniques*. 2001: 371-378.
- [4] Nimier-David M, Vicini D, Zeltner T, *et al.* Mitsuba 2: A retargetable forward and inverse renderer[J]. *ACM Transactions on Graphics (ToG)*, 2019, 38(6): 1-17.
- [5] Li T M, Aittala M, Durand F, *et al.* Differentiable monte carlo ray tracing through edge sampling[J]. *ACM Transactions on Graphics (TOG)*, 2018, 37(6): 1-11.
- [6] Beutel A, Mølhave T, Agarwal P K, *et al.* TerraNNI: natural neighbor interpolation on a 3D grid using a GPU[C]//*Proceedings of the 19th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*. 2011: 64-74.
- [7] Yariv L, Gu J, Kasten Y, *et al.* Volume rendering of neural implicit surfaces[J]. *Advances in neural information processing systems*, 2021, 34: 4805-4815.
- [8] Yuan Y J, Sun Y T, Lai Y K, *et al.* Nerf-editing: geometry editing of neural radiance fields[C]//*Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2022: 18353-18364.
- [9] Hu L, Zhang H, Zhang Y, *et al.* Gaussianavatar: Towards realistic human avatar modeling from a single video via animatable 3d gaussians[C]//*Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2024: 634-644.
- [10] Deng B, Yao Y, Dyke R M, *et al.* A Survey of Non-Rigid 3D Registration[C]//*Computer Graphics Forum*. 2022, 41(2): 559-589.
- [11] Chang Q, Wang W, Miyazaki J. Accelerating Nearest Neighbor Search in 3D Point Cloud Registration on GPUs[J]. *ACM Transactions on Architecture and Code Optimization*, 2025.
- [12] Chen Y, Chen Z, Zhang C, *et al.* Gaussianeditor: Swift and controllable 3d editing with gaussian splatting[C]//*Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024: 21476-21485.
- [13] Palandra F, Sanchietti A, Baieri D, *et al.* GSEdit: Efficient Text-Guided Editing of 3D Objects via Gaussian Splatting[J]. *arXiv preprint arXiv:2403.05154*, 2024.
- [14] Xiao H, Chen Y, Huang H, *et al.* Localized Gaussian Splatting Editing with Contextual Awareness[J]. *arXiv preprint arXiv:2408.00083*, 2024.
- [15] Chen Y, Chen A, Chen S, *et al.* Plasticine3D: Non-rigid 3D editing with text guidance[J]. *arXiv preprint arXiv:2312.10111*, 2023.
- [16] Lugmayr A, Danelljan M, Romero A, *et al.* Repaint: Inpainting using denoising diffusion probabilistic models[C]//*Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2022: 11461-11471.
- [17] Brooks T, Holynski A, Efros A A. Instructpix2pix: Learning to follow image editing instructions[C]//*Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2023: 18392-18402.
- [18] Xu T X, Hu W, Lai Y K, *et al.* Texture-gs: Disentangling the geometry and texture for 3d gaussian splatting editing[C]//*European Conference on Computer Vision*. Cham: Springer Nature Switzerland, 2024: 37-53.
- [19] Wu G, Yi T, Fang J, *et al.* 4d gaussian splatting for real-time dynamic scene rendering[C]//*Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024: 20310-20320.
- [20] Luiten J, Kopanas G, Leibe B, *et al.* Dynamic 3d gaussians: Tracking by persistent dynamic view synthesis[J]. *arXiv preprint arXiv:2308.09713*, 2023.

- [21] Shin I, Yu Q, Shen X, *et al.* Enhancing Temporal Consistency in Video Editing by Reconstructing Videos with 3D Gaussian Splatting[J]. arXiv preprint arXiv:2406.02541, 2024.
- [22] Sun Y T, Huang Y H, Ma L, *et al.* Splatter a Video: Video Gaussian Representation for Versatile Processing[J]. arXiv preprint arXiv:2406.13870, 2024.
- [23] Ji S, Wu G, Fang J, *et al.* Segment any 4d gaussians[J]. arXiv preprint arXiv:2407.04504, 2024.
- [24] Zhang Q, Xu Y, Wang C, *et al.* 3DitScene: Editing Any Scene via Language-guided Disentangled Gaussian Splatting[J]. arXiv preprint arXiv:2405.18424, 2024.
- [25] Radford A, Kim J W, Hallacy C, *et al.* Learning transferable visual models from natural language supervision[C]//International conference on machine learning. PmlR, 2021: 8748–8763.
- [26] Huang Y H, Sun Y T, Yang Z, *et al.* Sc-gs: Sparse-controlled gaussian splatting for editable dynamic scenes[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2024: 4220–4230.
- [27] Qu Y, Chen D, Li X, *et al.* Drag your gaussian: Effective drag-based editing with score distillation for 3d gaussian splatting[J]. arXiv preprint arXiv:2501.18672, 2025.
- [28] Yan J, Zhao A, Hu Y. Dragen3D: Multiview Geometry Consistent 3D Gaussian Generation with Drag-Based Control[J]. arXiv preprint arXiv:2502.16475, 2025.
- [29] Huang J, Yu H, Zhang J, *et al.* Point'n Move: Interactive scene object manipulation on Gaussian splatting radiance fields[J]. IET Image Processing, 2024, 18(12): 3507–3517.
- [30] Ye M, Danelljan M, Yu F, *et al.* Gaussian grouping: Segment and edit anything in 3d scenes[C]//European Conference on Computer Vision. Springer, Cham, 2025: 162–179.
- [31] Wang Y, Wu Q, Zhang G, *et al.* Learning 3D Geometry and Feature Consistent Gaussian Splatting for Object Removal[C]//European Conference on Computer Vision. Springer, Cham, 2025: 1–17.
- [32] Shen Q, Yang X, Wang X. Flashsplat: 2d to 3d gaussian splatting segmentation solved optimally[C]//European Conference on Computer Vision. Springer, Cham, 2025: 456–472.
- [33] Liu Z, Ouyang H, Wang Q, *et al.* InFusion: Inpainting 3D Gaussians via Learning Depth Completion from Diffusion Prior[J]. arXiv preprint arXiv:2404.11613, 2024.
- [34] Xu J, Wang Y, Zhao Y, *et al.* 3D StreetUnveiler with Semantic-Aware 2DGS[J]. arXiv preprint arXiv:2405.18416, 2024.
- [35] Yan Z, Li L, Shao Y, *et al.* 3dsceneeditor: Controllable 3d scene editing with gaussian splatting[J]. arXiv preprint arXiv:2412.01583, 2024.
- [36] Huang S Y, Chou Z T, Wang Y C F. 3d gaussian inpainting with depth-guided cross-view consistency[C]//Proceedings of the Computer Vision and Pattern Recognition Conference. 2025: 26704–26713.
- [37] Pardoel A M. Patch-Based Inpainting of 3D Gaussian Splatting[D]. Delft: Delft University of Technology, 2024.
- [38] Barnes C, Shechtman E, Finkelstein A, *et al.* PatchMatch: A randomized correspondence algorithm for structural image editing[J]. ACM Transactions on Graphics (TOG), 2009, 28(3): 24:1–24:11.
- [39] Jensen R, Dahl A, Vogiatzis G, *et al.* Large scale multi-view stereopsis evaluation[C]//Proceedings of the IEEE conference on computer vision and pattern recognition. 2014: 406–413.
- [40] Li T, Slavcheva M, Zollhoefer M, *et al.* Neural 3d video synthesis from multi-view video[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022: 5521–5531.
- [41] Gao Q, Wu Y, Wen C, *et al.* Relaygs: Reconstructing dynamic scenes with large-scale and complex motions via relay gaussians[J]. arXiv preprint arXiv:2412.02493, 2024.
- [42] Perazzi F, Pont-Tuset J, McWilliams B, *et al.* A benchmark dataset and evaluation methodology for video object segmentation[C]//Proceedings of the IEEE conference on computer vision and pattern recognition. 2016: 724–732.
- [43] Barron J T, Mildenhall B, Verbin D, *et al.* Mip-nerf 360: Unbounded anti-aliased neural radiance fields[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022: 5470–5479.
- [44] Ziegler I, Köksal A, Elliott D, *et al.* Craft your dataset: Task-specific synthetic dataset generation through corpus retrieval and augmentation[J]. arXiv preprint arXiv:2409.02098, 2024.
- [45] Deitke M, Schwenk D, Salvador J, *et al.* Objaverse: A universe of annotated 3d objects[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2023: 13142–13153.
- [46] Downs L, Francis A, Koenig N, *et al.* Google scanned objects: A high-quality dataset of 3d scanned household items[C]//2022 International Conference on Robotics and Automation (ICRA). IEEE, 2022: 2553–2560.
- [47] Mildenhall B, Srinivasan P P, Ortiz-Cayon R, *et al.* Local light field fusion: Practical view synthesis with prescriptive sampling guidelines[J]. ACM Transactions on Graphics (ToG), 2019, 38(4): 1–14.
- [48] Yeshwanth C, Liu Y C, Nießner M, *et al.* Scannet++: A high-fidelity dataset of 3d indoor scenes[C]//Proceedings of the IEEE/CVF International Conference on Computer Vision. 2023: 12–22.
- [49] Mirzaei A, Aumentado-Armstrong T, Derpanis K G, *et al.* Spin-nerf: Multiview segmentation and perceptual inpainting with neural radiance fields[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2023: 20669–20679.
- [50] Wang Q, Wang Z, Genova K, *et al.* Ibrnet: Learning multi-view image-based rendering[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2021: 4690–4699.
- [51] Knapitsch A, Park J, Zhou Q Y, *et al.* Tanks and temples: Benchmarking large-scale scene reconstruction[J]. ACM Transactions on Graphics (ToG), 2017, 36(4): 1–13.
- [52] Haque A, Tancik M, Efros A A, *et al.* Instruct-nerf2nerf: Editing 3d scenes with instructions[C]//Proceedings of the IEEE/CVF international conference on computer vision. 2023: 19740–19750.
- [53] Kerr J, Kim C M, Goldberg K, *et al.* Lerf: Language embedded radiance fields[C]//Proceedings of the IEEE/CVF international conference on computer vision. 2023: 19729–19739.
- [54] Ren Z, Agarwala A, Russell B, *et al.* Neural volumetric object selection[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2022: 6133–6142.
- [55] Mayer N, Ilg E, Haussler P, *et al.* A large dataset to train convolutional networks for disparity, optical flow, and scene flow estimation[C]//Proceedings of the IEEE conference on computer vision and pattern recognition. 2016: 4040–4048.
- [56] Sun P, Kretschmar H, Dotiwalla X, *et al.* Scalability in perception for autonomous driving: Waymo open dataset[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2020: 2446–2454.
- [57] Wang D, Zhang T, Abboud A, *et al.* Innerf360: Text-guided 3d-consistent object inpainting on 360-degree neural radiance fields[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024: 12677–12686.